

MODULE 01 · DOMAIN 1 - CLOUD CONCEPTS

Cloud Value and Economics

Why organisations move, and how the money changes shape when they do.

DOMAIN 1 of 4 EXAM WEIGHT 24% MODULE 1 of 26

Almost every Domain 1 question is really asking: **which named benefit is this?**

The six advantages of cloud computing

- ◆ **Trade capital expense for variable expense.** Pay for what you consume.
- ◆ **Benefit from massive economies of scale.** Aggregated demand lowers unit costs.
- ◆ **Stop guessing capacity.** Scale to demand instead of to a forecast.
- ◆ **Increase speed and agility.** Resources in minutes; experiments become cheap.
- ◆ **Stop spending money running data centres.** No racking, cooling or physical security.
- ◆ **Go global in minutes.** Deploy into other Regions without building anything.

ON THE EXAM

These six are the vocabulary the exam grades you against. Learn them as a set, not as ideas.

Capital expense or variable expense?

CAPITAL EXPENSE, ON PREMISES

Pay first, use later

- Large payment before any workload runs
- Sized for a peak you guessed at
- Depreciates whether used or idle
- Wrong guess means waste or an outage

VARIABLE EXPENSE, ON AWS

Use first, pay after

- Charged for actual consumption
- Sized to real demand, adjusted continuously
- Stops when the resource stops
- Wrong guess is corrected the same day

THE TIEBREAKER

Trading fixed for variable is not the same as eliminating cost. Any option saying cloud removes all expense is wrong.

Economies of scale, precisely

- ◆ **The mechanism is aggregation.** Hundreds of thousands of customers' usage combines into enormous purchasing volume.
- ◆ That volume lowers AWS's own unit cost for hardware, power and networking.
- ◆ AWS passes the reduction on as lower pay-as-you-go prices.
- ◆ It is not price matching against competitors.
- ◆ It is not the Free Tier, and it is not unlimited resources.

ON THE EXAM

If the question says aggregating usage across many customers, the answer is economies of scale. Every time.

Agility or elasticity?

AGILITY

Speed of doing something new

- Launch a campaign in three days, not three weeks
- Try an idea, discard it cheaply
- Measured in time to first result
- About the organisation's pace

ELASTICITY

Capacity following demand

- Scale out for Friday, scale in on Sunday
- Add and remove resources automatically
- Measured in capacity matching load
- About the workload's shape

THE TIEBREAKER

Agility answers how fast can we start. Elasticity answers how well does capacity track demand.

Benefit names and their giveaway phrases

IF THE SCENARIO MENTIONS	THE NAMED BENEFIT IS
Launching faster, experimenting cheaply	Agility, or increased speed and agility
Capacity rising and falling with load	Elasticity
Surviving failures, staying available, better security posture	Operational resilience
Staff freed from racking, patching and undifferentiated work	Staff productivity
Lower unit prices from aggregated demand	Economies of scale
Deploying to another Region without building anything	Global reach, or go global in minutes
No large up-front hardware purchase	Trade capital expense for variable expense

ON THE EXAM

Read the scenario looking for which of these seven phrases it is paraphrasing. That is the whole task.

Operational resilience and staff productivity

- ◆ **Operational resilience** covers availability, fault tolerance, disaster recovery and improved security controls, together.
- ◆ A company migrating for higher uptime and better security is describing operational resilience, not cost savings.
- ◆ **Staff productivity** means people stop doing undifferentiated heavy lifting and do business work instead.
- ◆ Racking servers, replacing failed disks and patching hypervisors are the classic examples of that heavy lifting.
- ◆ Neither of these means cost savings, even though both usually produce cost savings.

ON THE EXAM

Cost savings is a distractor for both. Pick it only when the scenario is explicitly about money.

Where on-premises money actually goes

01

Hardware

Servers, storage arrays, network gear, plus a refresh cycle.

02

Facilities

Space, power, cooling, physical security, fire suppression.

03

Licensing

Operating systems and databases, often per socket or per core.

04

Staffing

People to rack, cable, patch and replace it all.

05

Idle capacity

The headroom bought for a peak that arrives rarely.

ON THE EXAM

Migration eliminates data centre and hardware costs. It does not eliminate security, monitoring or training costs.

Bring your own licence, or licence included?

BYOL

You already own the licence

- Reuse an existing agreement on AWS
- Often needs Dedicated Hosts for socket-based terms
- Compliance tracking stays your job
- Cheaper if the licence is already sunk cost

LICENCE INCLUDED

AWS bundles it in the hourly rate

- Windows or SQL Server priced into the instance
- No separate vendor agreement to manage
- Stops costing when the instance stops
- Simpler, usually more expensive per hour

THE TIEBREAKER

Socket-bound or core-bound licences that must see the physical server are the classic reason to use Dedicated Hosts.

Rightsizing and automation

- ◆ **Rightsizing** means matching instance size to measured usage, continuously, rather than once at launch.
- ◆ AWS Compute Optimizer and Cost Explorer rightsizing recommendations are the named tools.
- ◆ **Automation** reduces cost by removing manual effort and by shutting resources off when idle.
- ◆ Scheduling non-production instances to stop overnight is the textbook example.
- ◆ Rightsizing is not the same as autoscaling. Rightsizing changes the size. Autoscaling changes the count.

ON THE EXAM

Size versus count is the distinction. Compute Optimizer suggests a different size; Auto Scaling adds another instance.

A retailer runs its own data centre. Every November it buys additional servers for the Christmas peak, and for the remaining eleven months roughly sixty percent of that hardware sits idle.

After migrating to AWS, the finance director notices the company no longer makes a large hardware purchase each autumn. Instead the monthly bill rises through November and falls again in January.

Which advantage of cloud computing does this most directly describe?

- | | |
|--|---|
| ☐ A Benefit from massive economies of scale | ☐ B Trade capital expense for variable expense |
| ☐ C Increase speed and agility | ☐ D Go global in minutes |

◆ PAUSE AND DECIDE

Which advantage of cloud computing does this most directly describe?

- A

 Benefit from massive economies of scale
True of AWS generally, but nothing in the scenario mentions aggregated demand or lower unit prices.
- B

 Trade capital expense for variable expense
The purchase disappeared and the bill now tracks usage. That is exactly this benefit.
- C

 Increase speed and agility
Tempting, but nothing here got faster. The timeline is identical, only the payment shape changed.
- D

 Go global in minutes
No second Region, no new geography. Nothing in the scenario is about reach.

THE SIGNAL

The autumn hardware purchase disappearing, replaced by a bill that follows demand.

A media company used to need six weeks to get servers approved, ordered, delivered and racked before it could test a new video format. Two of the last three formats were abandoned after testing.

On AWS the same test now takes two days to set up, and the team runs four experiments a quarter instead of one.

Which benefit of the AWS Cloud does this scenario demonstrate?

- ☐ A Elasticity
- ☐ B Operational resilience
- ☐ C Increased business agility
- ☐ D High availability

◆ PAUSE AND DECIDE

Which benefit of the AWS Cloud does this scenario demonstrate?

- A

 Elasticity
Nothing scales with demand here. The workload size never changes, only how quickly it can start.
- B

 Operational resilience
No failure, no outage, no security improvement is described.
- C

 Increased business agility
Six weeks became two days, and the experiment rate quadrupled. That is speed to act.
- D

 High availability
Availability is about staying up. Nothing here was ever down.

THE SIGNAL

Six weeks to two days, and four experiments a quarter instead of one. Both measure time to act.

A logistics firm is building its migration business case. It has listed everything it currently spends money on and wants to know which lines will actually disappear.

The list includes: the data centre lease, server hardware refresh, its security operations team, its monitoring tooling subscription, and the up-front purchase of storage arrays.

Which two costs will migrating to AWS eliminate? (Choose two.)

- ☐ A Data centre costs
- ☐ B Data security costs
- ☐ C Fixed hardware costs
- ☐ D Monitoring costs
- ☐ E Staff training costs

◆ PAUSE AND DECIDE

Which two costs will migrating to AWS eliminate? (Choose two.)

- A** Data centre costs

AWS owns and runs the facilities. The lease, power and cooling genuinely go away.
- B** Data security costs

Securing your data stays your responsibility under the shared responsibility model. It moves, it does not vanish.
- C** Fixed hardware costs

The up-front purchase becomes consumption-based charging. The fixed line disappears.
- D** Monitoring costs

You still monitor, and CloudWatch is a billed service. Cheaper, not eliminated.
- E** Staff training costs

Training usually increases during a migration. People need to learn AWS.

THE SIGNAL

Eliminate means gone entirely. Only facilities and up-front hardware qualify. Everything else changes shape.

A finance team reviews its AWS bill and finds twelve development instances that have run continuously for eight months. Monitoring shows peak CPU utilisation of nine percent, and nobody uses them outside working hours.

The team asks what to change. Development says the instances must keep their current configuration and data during working hours.

Which two actions address this most directly? (Choose two.)

- ☐ A Rightsize the instances to a smaller type
- ☐ B Put the instances behind an Auto Scaling group
- ☐ C Schedule the instances to stop outside working hours
- ☐ D Buy three-year Reserved Instances for them
- ☐ E Move the workload to Spot Instances

◆ PAUSE AND DECIDE

Which two actions address this most directly? (Choose two.)

- A

 Rightsize the instances to a smaller type
Nine percent peak utilisation is the textbook rightsizing signal. Smaller type, same configuration.
- B

 Put the instances behind an Auto Scaling group
Would work, but scales count when the problem is size and idle hours. More effort than needed.
- C

 Schedule the instances to stop outside working hours
Nobody uses them at night and stopping preserves configuration and data. Automation removing idle spend.
- D

 Buy three-year Reserved Instances for them
Commits three years to instances that are the wrong size and mostly idle. It discounts the waste.
- E

 Move the workload to Spot Instances
Interruption would destroy the developers' working state, which the scenario says must persist.

THE SIGNAL

Two planted constraints: nine percent utilisation points at rightsizing, and no out-of-hours use points at scheduling.

What to carry out of this module

- ◆ **The six advantages** are vocabulary. The exam grades the name, not your reasoning.
- ◆ **Agility is time. Elasticity is capacity.** Find which one the stem is measuring.
- ◆ **Economies of scale means aggregation** across customers. Not price matching, not the Free Tier.
- ◆ Migration eliminates facilities and up-front hardware. Security, monitoring and training only change shape.
- ◆ Rightsizing changes size. Auto Scaling changes count. Both save money, differently.

ON THE EXAM

When a Domain 1 question feels vague, ask which named benefit it is paraphrasing. There is nearly always exactly one.